Project Overview: Panther Cloud Air Airline Sim

Part 1: Timetable Generation System

Builds a flight system that:
- Creates flight schedules for the top 30 US Airports + NYC-Paris routes
- Assigns 55 aircrafts to routes
- Manages Gates, turnaround times, maintenance schedules
- Calculate realistic flight times (taxi, takeoff)
- Generates printable time tables
- Provides a passenger flight search interface

Part 2: 2 week flight simulation

Simulate 14 days of operations w/ various challenges

Day 1: Normal Operations
Day 3: 25% flights have weather delays
Day 5: Icing delays for flights above 40°N
Day 7: Strong jet stream
Day 9: N/A
Day 11: Aircraft failure at hub
Day 13: 8% flight west of 103°W cancelled
- Track passengers + flights + aircraft performance
- Generate financial reports (profit/loss)

Team Structure:
Project Leader - manages specs, designs
QA Manager - sees control and standards
Documentation Spec. - all docs + manuals
Principal Dev/Trainers - implementation and technical training

Key Calculations
1. Great Circle Distance between airports
2. Flight heading for wind adjustments
3. Gate to gate time
4. Passenger Demand Model (based on metro pop.)
5. Financial Calculations (fuel, leasing, airport fees)

Important Constraints:
- 120 GB disk space limit
- Only flights >150 miles
- Airports operate 5am-1am domestic
- 40-50 min turnaround times
- Maintenance after 200 flight hrs
- Gets availability
- No teleporting aircraft overnight
- Unique flight #'s

II keep a bound journal documenting all decisions, meetings, designs
Weekly checkins w/ professor

costs.h and costs.cpp Thu

Cost Calculation 1/15/2026

I've built the majority of the financial engines for the airline. This handles all the money calculations we need for the simulation.

- Currency conversion - converts between USD and EUR since we have Paris operations. xe.com rate exchange
- Fuel costs - calculates fuel expenses for both US flights (at $6.14 a gallon) and Paris flights (€1.97/liter converted to USD)
- Airport fees - handling the $2000 US takeoff fees and the €2100 Paris fees
- Flight operating costs - combines fuel w/ airport fees to get total cost per flight
- Aircraft leasing - calculates monthly lease costs for our fleet
- Ticket pricing - Prices tickets assuming 30% load factor, ensuring profitability
- revenue and profit - Tracks passenger revenue and calculates profit and loss

This gives us the financial reporting we need for at the end of the day (14-day simulation). Includes total passengers, operating costs, revenue, and profit and loss.

flight.h and flight.cpp Tuesday

Flight Engine Physics 01/20/2026

Finley and I implemented all of flight calcs from the project.

What makes it realistic is:

Distance calculations: uses the great-circle formula to get the true distance between airports based on their GPS coordinates

Heading/bearing: determines the compass direction between airports which we use for wind corrections

Cruising altitude: Selects the appropiate cruising altitude based on flight length, from about 20000 ft for short hops under 200 miles up to 38000 ft for the long-haul and international routes.

Wind adjustments: Applies a 9.5% time adjustment for east vs west flights to account for winds and Earths rotation

Fuel consumption: Estimates total fuel burn in gallons based on aircraft type and route distance

Also climb and descent, taxi times, and gate to gate times

From Project Doc: Page by Page

Page 4 (Paris Operations)
- Takeoff/landing fees: EUR 2100 per operation
- Fuel costs: EUR 1.97 per liter
- Exchange rate: Use xe.com on 1/31/2026

US Airport fees: 2000 per takeoff and landing

Page 6 Costs:
US fuel: $6.19 per gallon
Boeing 737-600 lease: $245000/month
Boeing 737-800: $270000
Airbus A220-100: $192000
Airbus A200-300: $228000

Page 7 Pricing Strategy:
Ticket Pricing: Assume 30% load factor to ensure profit
Goal: 75% load factor for actual operations

Constraints + Risks

So now we will be talking about the constraints and risks that we have come across for this project + the future

The first constraint is the exchange rate volatility between euros and US dollars. We always make sure that we keep a close eye on the conversion rate because it changes daily. This is where we check up on a specific currency converter to make sure its right.

The next constraint is time, and you can't argue that this is argobally the biggest constraint. We all have a hard deadline at the end of the semester. Thus, meaning that there are no extensions or extra weeks since were seniors. No room 4 delays!!

The third constraint is the memory, which is a 120 GB limit of disk space on the client machine. While it sounds like a lot when you are simming two weeks for over 50 aircraft flying between 31 airports, logging every flight and passenger, the database grows quickly w/o one even noticing.

Another constraint is between the client environment and our development environment. Only thing to say is having good communication mitigates that risk.

Three of our risks are data integration failure, time table generation complexity, and data accuracy.

The biggest pebble is database integration failure. The way we solve this so far is by making test files to make certain that everything is running smoothly at all steps.
No room for failure!

The second major risk is time table generation. Multiple constraints in the risk can be very complex, such as making sure that an aircraft can be at one place at a time, and flights must be over 150 miles apart. If there are 2 planes trying to use the same gate, then the sim fails.

The final risk of data accuracy is in our calculations. If any of the calculations are wrong, then the entire time table is off.

In summary, our constraints are manageable but require mitigation to avoid them.

+NFR

Functional Requirements

Starting with functional requirements. First, timetable generation The system automatically generates flight schedules for all 55 aircrafts. One of the key constraints is that routes to be more than 150 miles apart. The system calculates accurate gate-gate flight times which means accounting for taxi times, takeoff, climb, descent, and landings.

Second, Hub Optimization. We need to identify 3 or 4 optimal hub airports based on passenger flow and geography. A big priority is minimizing overnight layovers in New York for passengers connecting to Paris. Nobody wants to sleep in JFK when they are trying to get to Paris. And the hub network has to support all to all connectivity which is how major airlines operate. This is how we're building the same connectivity from day 1 as we can get from any airport to another airport through the hubs w/ at most 1 connection

Third, aircraft management. every aircraft needs scheduled maintenance every 200 flight hours, which takes about a day and a half at a hub. But here's the catch, we can only have 3 aircrafts in maintenance at any given hub at the same time which is because of limited maintenance bays. So the system has to intelligentally manage that scheduling conflict.

Fourth financial operations as this is a business so making money matters
Tickets priced at 30% load factor
Generates comphensive profit reports
Handles USD/EUR conversion
75% load makes profit

Functional- what the system does

Non-Functional Requirements

First, accuracy/precision